

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2022/2023

MAY/JUNE EXAMINATION

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

TUESDAY, 30 MAY 2023

TIME: 6.30 PM – 8.30 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN COMPUTER SCIENCE (DATA SCIENCE)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) Artificial Intelligence (A.I.) is one of the newest disciplines, formally initiated in 1956 when the name was coined. Over the last few decades, hundreds of definitions of A.I. have surfaced and these definitions can be distinguished along two main dimensions, resulting in four approaches for developing intelligent systems.
- Draw a diagram with appropriate labels to illustrate the classification of the **FOUR (4)** approaches mentioned above. (5 marks)
 - The *Turing Test* was proposed to test a machine's ability to demonstrate human-like intelligence. As A.I. technology improved, many of its variations were proposed, which include the *Total Turing Test*. Briefly describe **TWO (2)** additional requirements of the *Total Turing Test*. (2 marks)
- b) *Minesweeper* is a game that requires one to locate the mines hidden in a grid of squares without being blown up. Safe squares have numbers that indicate how many mines are surrounding the square. These numbers serve as a clue to solving the game by opening all of the other safe squares. One can mark the predicted location of a mine with a flag. However, if one clicks on a mine, this will result in losing the game! **Figure 1** shows a sample gameplay of the game.

**Figure 1. Minesweeper game**

As a *Minesweeper* lover, you plan to develop a *rational agent* that solves the *Minesweeper* game.

- Identify the *environment* where the *Minesweeper* agent is going to operate. (1 mark)
- Identify **SEVEN (7)** properties of the *environment* mentioned above. Justify your answer. (7+7 marks)
- "The *Minesweeper* agent is an omniscient agent". State your agreement with this statement. Justify your answer. (1 + 2 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 2**

The *Travelling Salesman Problem (TSP)* is a serious challenge for the logistics and supply chain industry. Given a set of cities and the distance between every pair of cities, the problem is to find the shortest possible route that visits every city exactly once and then returns back to the starting point.

- a) “*TSP is an optimization problem*”.
State your agreement with this statement. Justify your answer. (1 + 2 marks)
- b) Explain **TWO (2)** ways how classical search is different from local search. (4 marks)
- c) *TSP* can be solved by *A* search*, which is one of the most promising classical search strategies.
- (i) Evaluate *A* search*'s overall performance in terms of completeness, cost-optimality, time complexity and space complexity. (4 marks)
 - (ii) Identify **TWO (2)** factors that affect its cost-optimality. (2 marks)
- d) *TSP* can also be solved by *random-restart hill climbing search*, which is a local search strategy. With the use of appropriate labels, draw a *hill-climbing landscape* that describes various states *random-restart hill climbing search* would encounter. It is important to note that the landscape's elevation corresponds to an objective function. (8 marks)
- e) Which strategy is more suitable in solving *TSP*, *A* search* or *random-restart hill climbing search*? Justify your answer. (1 + 3 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 3**

a) Games have long been a popular area for research in Artificial Intelligence (A.I.). They are challenging and yet easy to formalise. Hence, they are used as platforms for the development of new A.I. methods and for measuring how well they work. The most commonly studied games in A.I. are *turn-taking, two-player, perfect information, deterministic, and zero-sum*.

(i) Provide an example of the most commonly studied game in A.I. (1 mark)

(ii) The game provided in **Question 3 a) (i)** can be formally defined with the following six elements. Briefly describe these **SIX (6)** elements. (6 marks)

- S_0
- $ACTIONS(s)$
- $RESULT(s, a)$
- $TO-MOVE(s)$
- $IS-TERMINAL(s)$
- $UTILITY(s, p)$

b) A *game tree* is a type of search tree that represents all possible game states as a result of each move made by two players, *MAX* and *MIN*, alternatively within a game. Answer the following **FOUR (4)** questions based on the game tree provided in **Figure 2**.

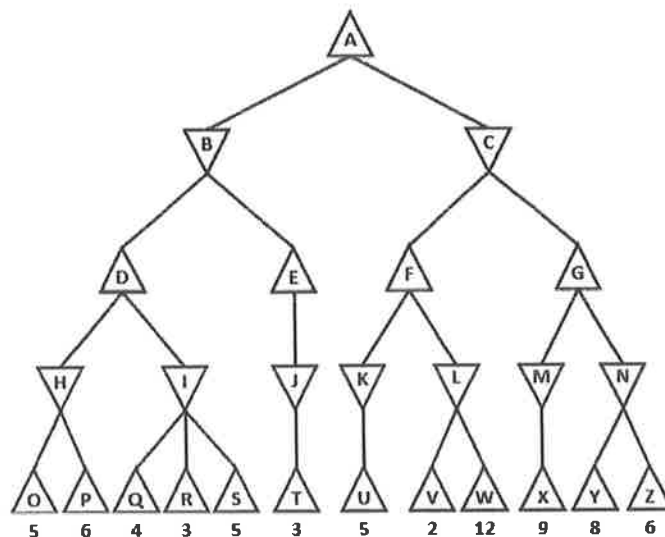


Figure 2. Game tree

(i) Compute the *minimax value* for each node, from node A to node N. (7 marks)

(ii) Identify the best move for the *MAX* player at the root node. (1 mark)

(iii) Apply *minimax search* to the game tree with *alpha-beta pruning* in *left-to-right order* to deduce nodes that would be pruned. (8 marks)

(iv) “Many tree branches can be pruned away with *alpha-beta pruning*, but its effectiveness is not always guaranteed”.

State your agreement with this statement. Justify your answer.

(1 + 1 marks)

[Total: 25 marks]

Question 4

Sudoku is a logic-based puzzle game. In its simplest configuration, a *Sudoku* can consist of a 4×4 grid with numbers appearing in some of the squares, as shown in **Figure 3**. The objective of this game is to fill the 4×4 grid with digits so that each column, each row, and each of the four 2×2 sub-grids (highlighted in bold lines) that compose the grid contain all of the digits from 1 to 4.

	1	2	3	4
A	2			
B		1	3	
C	3			1
D		2	4	

Figure 3. 4×4 Sudoku

- a) Formulate the *Sudoku* given in **Figure 3** as a *Constraint Satisfaction Problem (CSP)* with the use of the *Alldiff* constraint. (9 marks)
- b) Explain how *Alldiff* can be implemented with the use of simple conditions checking to detect any violation of constraint. (3 marks)
- c) Suppose all *Alldiff* constraints have been expanded into binary constraints and *AC-3* algorithm is applied, explain how the following variables are made *arc-consistent* step-by-step.
 - (i) *B1* (3 marks)
 - (ii) *D1* (2 marks)
 - (iii) *A2* (2 marks)
 - (iv) *C2* (2 marks)
- d) If inference is applied onto all squares without numbers in **Figure 3**, eventually, the entire *Sudoku* will be solved by *AC-3*. Name **TWO (2)** properties of the solution. Then, briefly describe each of them. (2 + 2 marks)

[Total: 25 marks]